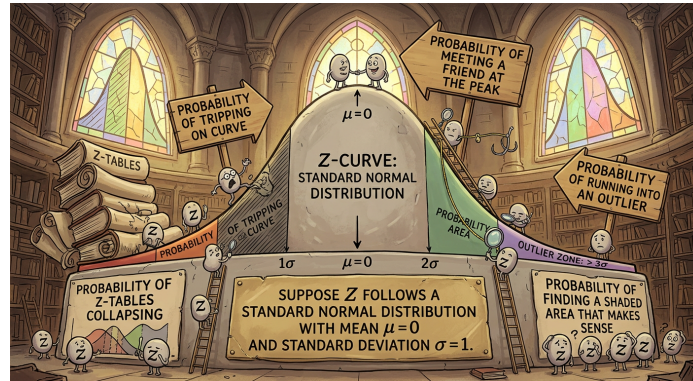


Unit 06: Normal Distribution & CTL

Instructor: Dr. Md Mutasim Billah

Example 1: Standard Normal Probabilities



Suppose Z follows a standard normal distribution with mean $\mu = 0$ and standard deviation $\sigma = 1$. Find the following probabilities.

(a) $P(Z < -0.77)$

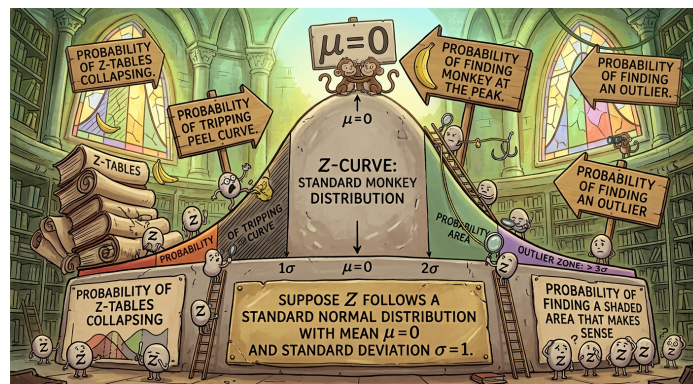
(b) $P(Z > -0.77)$

(c) $P(-1.05 < Z < 2.41)$

(d) Find each of the following:

$$P(-1 < Z < 1), \quad P(-2 < Z < 2), \quad P(-3 < Z < 3).$$

Example 2: Finding Standard Normal Cutoffs



Suppose Z follows a standard normal distribution with mean $\mu = 0$ and standard deviation $\sigma = 1$. Find the value of z in each case.

(a) $P(Z < z) = 0.0708$

(b) $P(Z > z) = 0.6480$

(c) $P(-z < Z < z) = 0.95$, where $z > 0$.

(d) $P(-1.21 < Z < z) = 0.7330$

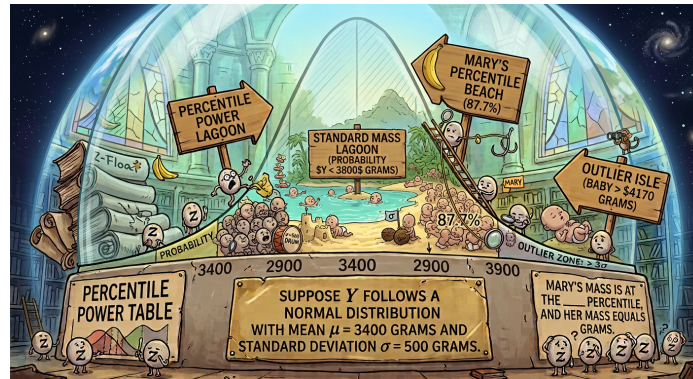
Example 3: Standard Normal Percentiles

Suppose Z follows a standard normal distribution with mean $\mu = 0$ and standard deviation $\sigma = 1$. Answer the following questions.

(a) What are the 90th and 10th percentiles of Z ?

(b) What are the 75th and 25th percentiles of Z ?

Example 4: Newborn Mass



Let Y be the mass, in grams, of a randomly selected newborn baby. Assume that Y follows a normal distribution with mean $\mu = 3400$ grams and standard deviation $\sigma = 500$ grams.

- (a) What is the probability that a randomly selected newborn baby has a mass greater than 4170 grams?
- (b) What is the probability that a randomly selected newborn baby has a mass less than 3800 grams?

- (c) 87.7% of all newborn babies are lighter than Mary. Mary's mass is at the _____ percentile, and her mass equals _____ grams.

Example 6: Filling Milk Jugs



Let Y be the volume of milk, in gallons, placed in a plastic jug. Assume that Y follows a normal distribution with mean $\mu = 1.02$ gallons and standard deviation $\sigma = 0.01$ gallons.

- (a) Find the probability that a randomly selected jug is underfilled. Treat a jug as underfilled if it contains less than 1 gallon of milk.

- (b) Find the interval (A, B) centered at $\mu = 1.02$ that contains the milk volumes of 80% of all jugs. That is, find A and B such that

$$P(A < Y < B) = 0.80 \quad \text{and} \quad \frac{A + B}{2} = 1.02.$$

Example 7: Sample Mean of Milk Volumes

Let Y be the volume of milk, in gallons, placed in a plastic jug. Assume that Y follows a normal distribution with mean $\mu = 1.02$ gallons and standard deviation $\sigma = 0.01$ gallons.

Assume that the milk volumes in independently selected jugs are independent observations from this distribution.

- (a) Suppose we randomly select $n = 2$ jugs and compute their sample mean, $\bar{Y}$. Find $P(\bar{Y} < 1)$.

- (b) What happens to $P(\bar{Y} < 1)$ as the sample size increases? Explain your reasoning.

Example 8: Sample Mean of Car-Part Weights



Let Y be the weight, in grams, of a randomly selected car part. Assume that $Y \sim \text{Uniform}(2.97, 3.01)$. Also, assume that the weights of the selected parts are independent.

- (a) Suppose we randomly select $n = 30$ car parts and compute their sample mean, $\bar{Y}$. Find $P(2.987 < \bar{Y} < 2.991)$.

- (b) Suppose we randomly select $n = 50$ car parts and compute their sample mean, $\bar{Y}$. Find

$$P(2.987 < \bar{Y} < 2.991).$$